Asu Singh

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Summary

Software Development Engineer with 4+ years of experience in financial services, specializing in Java, multithreading, and database optimization. Experience with Java 8+, Spring Boot, microservices, REST APIs, Hibernate, relational databases, test-driven development, and production-performance improvements.

Experience

Software Development Engineer - ION Group

Noida, India | Aug 2020-Present

- Designed and developed XTP, ION's real-time back-office trading product.
- Worked with XTP's in-house MQ-like event-driven notification system, understanding its internal workings and operational failure points.
- Led a component-refactoring initiative using Java multithreading, increasing performance by 20x.
- Developed Alazium, a log-analysis initiative that reduced production issue-analysis time by 30%.
- Designed a recovery feature for halted end-of-day processing, reducing downtime by 90%.
- Led migration of XTP CI systems from Red Hat 7 to Red Hat 8.
- Redesigned historical-data cleanup using a tweaked linear-regression approach, improving database performance and reducing processing time by 95% under load.

Software Development Intern - Holisol Logistics

Delhi, India | Jun 2019-Sep 2019

- Developed a Daily Run Sheet Generator with a Django frontend to optimize deliveries.
- Presented the application to the company founder and secured deployment approval.

Projects

Hybrid Social Group Algorithm

- Developed a meta-heuristic algorithm to overcome local minima and maxima in machine-learning processes.
- Applied the algorithm to COVID-19 detection from chest X-ray images, achieving 99.31% accuracy.
- Published the work in *Cognitive Computation*; it has been cited by several publications.

Lung Disease Prediction

- Developed an automated application to detect and predict the onset of lung diseases from X-ray images.
- Achieved over 99% accuracy using a meta-heuristic algorithm.

Publications

COVID-19 Infection Detection from Chest X-Ray Images

Springer | Mar 2021

- Developed the Hybrid Social Group Optimization algorithm for COVID-19 detection from chest X-ray images with over 99% accuracy.
- Published in the *Cognitive Computation* journal.

Education

Bachelor of Technology

Maharaja Agrasen Institute of Technology, Delhi

Skills

- Languages: Java, Python, C++, C#
- Backend & data: Java 8+, Spring Boot, microservices, REST APIs, Hibernate, event-driven systems, MQ-like messaging, Oracle, PostgreSQL, Django
- Engineering: Multithreading, performance tuning, test-driven development, JUnit, Mockito, Agile, Unix
- Cloud & observability: AWS, Grafana, Prometheus
- Frontend: HTML5, CSS, JavaScript
- Other: Machine learning

Awards

- Top 0.01% in 10th standard (Sep 2013); awarded by the HRD Minister and invited to the INSPIRE Program.
- Top 0.1% performer at ION Group.

- Top 0.01% in JEE Advanced 2016.
- Top 10, inaugural HackBMU Hackathon, for developing an Illegal Mining Detector.